

The Three-Nighbor Percolation Number of the 8×8 Torus

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

Abstract

For the three-neighbor bootstrap process on the Cartesian-product torus $C_8 \square C_8$, we prove that the minimum size of a percolating set is 22. The lower bound follows by identifying the process on the initially uninfected vertices with parallel leaf stripping and then counting the edges of the resulting forest. For the upper bound we give a labelled 22-vertex set and its complete exact synchronous infection trace.

1 Statement and context

Write $V = \mathbb{Z}_8 \times \mathbb{Z}_8$, with two vertices adjacent when they differ by 1 modulo 8 in exactly one coordinate. From an initially infected set $S \subseteq V$, define

$$I_0 = S, \quad I_{r+1} = I_r \cup \{v \in V \setminus I_r : |N(v) \cap I_r| \geq 3\}.$$

Let $t_3(8, 8)$ be the minimum cardinality of an S for which $\bigcup_{r \geq 0} I_r = V$.

Minimum percolating sets on grids and tori have been considered in several closely related formulations; see, for example, Benevides, Bermond, Lesfari, and Nisse [1] and Flocchini, Lodi, Luccio, Pagli, and Santoro [3]. Most directly, Bushaw and Clifton [2, Theorems 1.6–1.8] record general toroidal bounds which, for this instance, give

$$22 \leq t_3(8, 8) \leq 23.$$

Their Question 4.3 asks when the lower bound is strict and identifies the 8×10 torus as a strict case; it does not determine this fixed 8×8 case. We close only this one-unit gap and make no assertion about other dimensions or about priority beyond this comparison.

Theorem 1. *For the synchronous three-neighbor process on $C_8 \square C_8$,*

$$t_3(8, 8) = 22.$$

2 Leaf stripping and the lower bound

Lemma 2. *Let G be any finite 4-regular graph, let S be initially infected, and put $U = V(G) \setminus S$. The three-neighbor process from S percolates if and only if the induced graph $G[U]$ is a forest. Moreover, the synchronous infection layers outside S are exactly the parallel degree-at-most-one deletion layers of $G[U]$.*

Proof. If R is the set still uninfected at the start of a round, then a vertex $v \in R$ has at least three infected neighbors precisely when $\deg_{G[R]}(v) \leq 1$. Thus a round of infection is simultaneous deletion of all vertices of degree at most one from $G[R]$. If $G[U]$ contains a cycle, every vertex of that cycle retains its two cycle neighbors and the deletion process cannot empty U . Conversely, repeated deletion of all vertices of degree at most one empties every finite forest. The assertion about layers follows from the same equivalence at each round. $\square$

Lemma 3. *Every percolating set in $C_8 \square C_8$ has at least 22 vertices.*

Proof. Let S percolate, put $k = |S|$ and $u = 64 - k$, and first suppose $u > 0$. By Lemma 2, $G[U]$ is a forest. If it has $c \geq 1$ components, then it has $u - c$ edges. Let b be the number of edges between U and S . Counting degrees on U gives

$$4u = 2(u - c) + b, \quad \text{so} \quad b = 2u + 2c.$$

Counting degrees on S , and writing $e(S)$ for the number of edges induced by S , gives

$$4k = 2e(S) + b \geq b.$$

Consequently

$$4k \geq 2u + 2c \geq 2(64 - k) + 2,$$

so $6k \geq 130$ and hence $k \geq 22$. If $u = 0$, then $k = 64$ and the conclusion is immediate. $\square$

3 A labelled witness and exact trace

All coordinates in what follows are represented by $0, \dots, 7$ and are reduced modulo 8. Take

$$S = \{(0, 0), (0, 1), (0, 5), (1, 2), (1, 4), (1, 7), (2, 0), (2, 5), (3, 2), (3, 4), (3, 7), \\ (4, 1), (4, 3), (4, 6), (5, 0), (5, 5), (6, 1), (6, 3), (6, 7), (7, 2), (7, 4), (7, 6)\}.$$

The set has 22 vertices. Table 1 lists every subsequent synchronous infection layer $L_r = I_r \setminus I_{r-1}$, with $L_0 = S$.

Table 1: The exact synchronous infection trace after $L_0 = S$.

Layer	Vertices
L_1	$\{(0, 2), (0, 4), (1, 0), (1, 5), (2, 4), (2, 7), (3, 3), (4, 2), (5, 1), (6, 0), (6, 2), (7, 1), (7, 3), (7, 5)\}$
L_2	$\{(0, 3), (1, 1), (5, 2), (7, 0)\}$
L_3	$\{(1, 3), (5, 3), (7, 7)\}$
L_4	$\{(0, 7), (2, 3)\}$
L_5	$\{(0, 6), (2, 2)\}$
L_6	$\{(1, 6), (2, 1)\}$
L_7	$\{(2, 6), (3, 1)\}$
L_8	$\{(3, 0), (3, 6)\}$
L_9	$\{(3, 5), (4, 0)\}$
L_{10}	$\{(4, 5), (4, 7)\}$
L_{11}	$\{(4, 4), (5, 7)\}$
L_{12}	$\{(5, 4), (5, 6)\}$

Layer	Vertices
L_{13}	$\{(6, 4), (6, 6)\}$
L_{14}	$\{(6, 5)\}$

For completeness, this table is an exact trace, not merely a legal infection ordering. Direct enumeration of the four modular neighbors gives, for every $1 \leq r \leq 14$,

$$L_r = \{v \in V \setminus I_{r-1} : |N(v) \cap I_{r-1}| \geq 3\}.$$

The layer sizes are

$$22, 14, 4, 3, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1,$$

so the disjoint layers contain all 64 vertices. Thus S percolates and $t_3(8, 8) \leq 22$. Lemma 3 supplies the reverse inequality and proves Theorem 1.

The coordinate data are also supplied in a machine-readable JSON certificate and a dependency-free checker. The checker reconstructs the torus, verifies the equality defining every displayed layer, and confirms that the 42-vertex complement of S is connected with 41 induced edges, hence is a tree. This computation is a reproducibility check for the finite trace; the lower-bound proof is independent of it.

Computational and AI assistance disclosure

The displayed witness was located using a mixed-integer cycle-cut search. Solver status is not used as proof. A separate exact checker, using only integer coordinates and finite sets, verifies the serialized witness and all synchronous layers. A language model assisted with proof review, exact-data checking, and manuscript preparation. The mathematical conclusion rests on the self-contained counting argument and the explicitly reproducible finite certificate above.

References

- [1] Fabricio Benevides, Jean-Claude Bermond, Hicham Lesfari, and Nicolas Nisse. Minimum lethal sets in grids and tori under 3-neighbour bootstrap percolation. *European Journal of Combinatorics*, 119:103801, 2024.
- [2] Neal Bushaw and Alexander Clifton. 3-neighbor bootstrap percolation on two-dimensional grids, 2026. arXiv:2608.06133v1.
- [3] Paola Flocchini, Elena Lodi, Fabrizio Luccio, Linda Pagli, and Nicola Santoro. Dynamic monopolies in tori. *Discrete Applied Mathematics*, 137(2):197–212, 2004.